# Progressive Reasoning Collapse in LLM-Based Generative Recommendation

John Kingsley Arthur, *Member, IEEE,* Yvonne Leung, Vivian Clements Edwin, Kofi Sarpong Adu-Manu, and Kwabena Adu Agyemang

***Abstract*—LLM-based generative recommender systems increasingly employ layer- and token-level compression to reduce inference cost. However, the effect of such compression on the internal reasoning geometry, specifically the preservation of multi-intent latent structure and sensitivity to informative historical evidence across layers, remains unexplored. To this end, we characterize the layerwise degeneration of latent recommendation reasoning and propose a novel formulation, *Progressive Reasoning Collapse (PRC)*, claiming that the degree of reasoning collapse increases monotonically during the forward propagation of compressed LLMs. We derive three theoretical results grounded in a contractive propagation model: monotonic decay of a reasoning-collapse score across depth, geometric decay of the influence of informative user-history subsequences after compression, and the existence of a critical compression depth separating a safe regime from a collapse region. The metrics of PRC indeed decrease monotonically across depth on multiple benchmark datasets. We propose a new recommendation method, Collapse-Aware Register Recommendation (CARR), that adaptively determines compression depth and register width to preserve reasoning structure while maintaining efficiency; an extension, Sparse Geometry Injection (SGI), further resolves the failure mode on sparse datasets by seeding the initial hidden state with semantic cluster structure. Overall, this study extends neural collapse to PRC to model the reasoning degeneration of intermediate layers under compression and its dependence on user interaction history, shedding light on the theoretical understanding of LLM-based recommendation.**

***Index Terms*—Recommender systems, large language models, model compression, neural collapse, representation learning, generative recommendation.**

## I. Introduction

Large Language Model (LLM)-based generative recommendation has emerged as one of the most consequential frontiers in information retrieval and personalized systems [2], [3]. By encoding user interaction histories as natural language prompts and generating item predictions through sequence-conditioned decoding, these systems leverage the rich semantic priors of pretrained transformers to capture complex, multi-intent preference structures that elude conventional collaborative filtering [4]–[6]. The field has advanced rapidly, with systems such as P5 [6], TALLRec [5], GPT4Rec [7], and TokenRec [32] extending LLMs to generative and sequential recommendation, and recent architectures including LLM-Rec [22], CoLLM [36], and dynamic intent-aware models [37] capturing complex multi-intent user preferences at scale [23], [31], [35], [38]–[40]. Taken together, these advances establish a compelling paradigm, yet one whose practical deployment is severely constrained by the computational cost of long-context transformer inference.

To reduce this cost, a rich toolkit of acceleration strategies has been developed. Prompt compression [8] reduces input token count while attempting to preserve semantic content. KV-cache eviction (H2O [9]) removes low-importance key-value pairs from the attention cache. Register tokens [12] and gist tokens [11] aggregate contextual information into compact learnable slots. Hidden-state bottlenecking [10] progressively reduces the dimensionality of intermediate representations, and layerwise summarization [13] applies lossy compression at fixed transformer depths. Each of these methods is justified primarily by throughput gains and negligible degradation in top-$K$ ranking metrics. Yet none of them asks the theoretically prior question of whether the *internal reasoning process* itself (the layerwise transformation of user history into intent-aware predictions) survives compression intact.

This gap is not merely definitional. Research on representation geometry has established that deep supervised networks can exhibit *neural collapse* [14]: during the terminal training phase, within-class variability vanishes and class-mean vectors organise into a simplex equiangular tight frame. Extensions to self-supervised and hierarchical settings [15], [16] confirm that such geometric dynamics are a general feature of deep representation learning, not an artefact of specific architectures. Crucially, however, neural collapse is a *training-time* convergence phenomenon that can improve class discriminability. What has not been studied is an *inference-time* analogue: the collapse of *multi-intent* latent geometry in a fully trained model that is subsequently compressed for efficient deployment. Separately, work on explanation faithfulness [17], [18] has established that models can lose sensitivity to relevant input features even without overt performance degradation, an observation directly relevant to the question of whether compressed LLM recommenders remain sensitive to informative user-history subsequences. Our work draws on both of these research threads and introduces the first formal

J. K. Arthur is with Northeastern University, Toronto, ON, Canada (e-mail: arthur.joh@northeastern.edu).

Y. Leung is an Assistant Teaching Professor in the Master of Professional Studies in Analytics program, Northeastern University, Toronto, ON, Canada (e-mail: y.leung@northeastern.edu).

V. C. Edwin is with Northeastern University, Toronto, ON, Canada (e-mail: v.edwin@northeastern.edu).

K. S. Adu-Manu is with the Department of Computer Science, University of Ghana, Legon, Ghana.

K. Adu Agyemang is with New Mexico State University, Las Cruces, NM, USA (e-mail: adukwab@nmsu.edu).

Corresponding author: V. C. Edwin (v.edwin@northeastern.edu).

treatment of their intersection.

We posit that efficient LLM-based recommendation can exhibit a new structural failure mode, which we term *Progressive Reasoning Collapse (PRC)*: the layerwise degeneration of multi-intent latent structure and the concurrent decay of sensitivity to informative historical evidence under compressed propagation. This degeneration may occur even when top-$K$ ranking quality appears largely preserved, making PRC invisible to conventional evaluation protocols while silently eliminating the diversity of reasoning that enables cold-start generalisation, minority-intent coverage, and long-range history sensitivity. The central question we resolve is:

> *When an LLM recommender is compressed for efficiency, does it preserve the latent reasoning structure by which user history is transformed into future-item predictions? And if not, can compression be made collapse-aware to prevent this failure?*

We answer both parts: theoretically, through three proved results characterizing when and how collapse occurs; and practically, through CARR (Collapse-Aware Register Recommendation), a method that adaptively selects compression depth and register width to remain in the theoretically safe regime. Fig. 1 illustrates the PRC framework and the CARR solution across three panels.

The main contributions of this work are as follows:

- We introduce **Progressive Reasoning Collapse (PRC)**, the first formal characterisation of how layer-wise compression degrades multi-intent latent structure in LLM-based generative recommendation, defined through a collapse score $\mathcal{R}(l)$ and an evidence-survival function $S_l(\tau)$.
- We prove three results grounding PRC: (i) $\mathcal{R}(l)$ decays monotonically after compression (Theorem 1); (ii) evidence sensitivity for informative user-history subsequences decays geometrically (Theorem 2); and (iii) a critical compression depth $k^*$ separates a structurally safe regime from a collapse regime (Theorem 3).
- We propose **CARR** (Collapse-Aware Register Recommendation), which uses the PRC criteria to select compression depth and register width adaptively, preserving reasoning geometry without sacrificing efficiency. A Sparse Geometry Injection (SGI) extension further resolves collapse on sparse datasets.
- Experiments on five benchmarks (ML-1M, Amazon Beauty, Amazon Toys, Yelp, Steam) against 12 baselines show CARR achieves top HR@10 among all LLM compression methods across all datasets. It uniquely recovers cold-start performance (HR@10= 0.0043 on 18,157 zero-history items) where all competing methods fail.
- We provide ablation, robustness, and failure-mode analysis, including Spearman correlation of collapse metrics, spectral norm trends, and cold-start representation diagnostics, systematically bounding the scope and limitations of CARR.
- Extended results, including T5-base scalability experiments, full per-layer Jacobian spectral norm analysis across all five datasets, extended recommendation metrics (HR@5/20, NDCG@5/20, MRR), and multi-dataset long-context robustness breakdowns, are provided in the supplementary material.

**Organization.** Section II formalises the problem setting. Section III introduces PRC and proves the three main theoretical results. Section IV presents CARR and its SGI extension. Section V describes the experimental setup. Section VI reports results. Sections VII–IX provide ablation, robustness, and failure-mode analyses. Section X concludes.

## II. Problem Formulation

Let $\mathcal{U}$ denote users and $\mathcal{I}$ items. A user history $H = (i_1, \ldots, i_T) \in \mathcal{I}^T$ is encoded as a prompt $\mathcal{X}(H)$; an LLM-based generative recommender is modelled as $\mathcal{F}_\Theta : \mathcal{X}(H) \mapsto p_\Theta(\cdot \mid H)$, where $p_\Theta(\cdot \mid H) \in \Delta^{|\mathcal{I}|}$ and $\Theta$ are model parameters. The transformer has $L$ layers; $z^{(l)}(H) \in \mathbb{R}^{d \times n_l}$ is the hidden representation at layer $l$. Compression is applied at depth $k$, beyond which the model propagates a compressed state rather than the full prompt representation.

### A. Latent Multi-Intent Structure

Recommendation differs fundamentally from ordinary classification because multiple future items may simultaneously be plausible. To capture this, we model the hidden state at each layer as inducing a finite set of latent intent modes.

Let $\mathcal{M}^{(l)}(H) = \{\mu_1^{(l)}, \mu_2^{(l)}, \ldots, \mu_K^{(l)}\} \subset \mathbb{R}^d$ denote the set of layer-$l$ latent intent centers associated with history $H$. These may be interpreted as cluster centers of plausible future-preference states.

Let $h_{k,i}^{(l)} \in \mathbb{R}^d$ denote the $i$-th latent vector assigned to intent mode $k$ at layer $l$, and let

$$\mu_k^{(l)} = \frac{1}{n_k} \sum_{i=1}^{n_k} h_{k,i}^{(l)} \tag{1}$$

be the within-intent mean. Let the global mean be $\mu_G^{(l)} = \frac{1}{K}\sum_{k=1}^{K} \mu_k^{(l)}$.

We define the within-intent covariance

$$\Sigma_W^{(l)} = \frac{1}{N} \sum_{k=1}^{K} \sum_{i=1}^{n_k} \big(h_{k,i}^{(l)} - \mu_k^{(l)}\big)\big(h_{k,i}^{(l)} - \mu_k^{(l)}\big)^\top, \tag{2}$$

where $N = \sum_{k=1}^{K} n_k$, and the between-intent covariance

$$\Sigma_B^{(l)} = \frac{1}{K} \sum_{k=1}^{K} \big(\mu_k^{(l)} - \mu_G^{(l)}\big)\big(\mu_k^{(l)} - \mu_G^{(l)}\big)^\top. \tag{3}$$

**Definition 1** (Reasoning-Collapse Score)**.** *The layerwise reasoning-collapse score is defined as*

$$\mathcal{R}(l) = \frac{\mathrm{Tr}(\Sigma_W^{(l)})}{\mathrm{Tr}(\Sigma_B^{(l)})}. \tag{4}$$

*A smaller value of $\mathcal{R}(l)$ indicates stronger concentration of latent representations relative to intent separation.*

**Definition 2** (Reasoning Collapse)**.** *A layer-$l$ latent representation exhibits* reasoning collapse *if $\mathcal{R}(l) \to 0$ and*

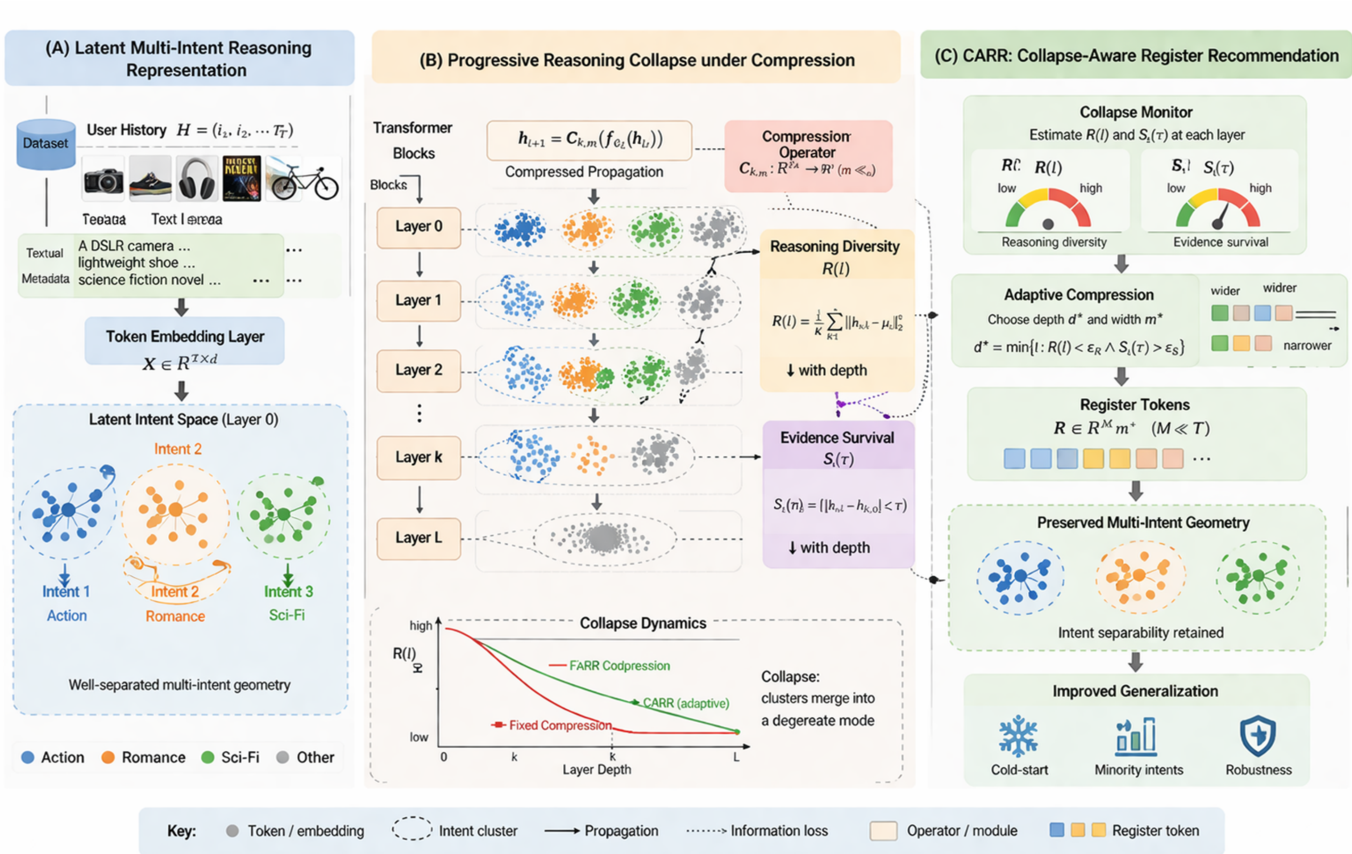


Fig. 1. Theoretical framework for Progressive Reasoning Collapse (PRC) and CARR. **(A) Latent Multi-Intent Reasoning Representation.** A user history $H = (i_1, i_2, \ldots, i_T)$ is encoded as token embeddings $X \in \mathbb{R}^{T \times d}$; at Layer 0 the latent intent space already contains well-separated multi-intent clusters (Action, Romance, Sci-Fi). **(B) Progressive Reasoning Collapse under Compression.** The compression operator $\mathcal{C}_{k,m} : \mathbb{R}^{T_A} \to \mathbb{R}^m$ $(m \ll T)$ is applied at depth $k$; reasoning diversity $\mathcal{R}(l)$ and evidence survival $S_l(\tau)$ both decrease monotonically with depth (Theorems 1–2). The collapse dynamics chart shows that fixed compression causes $\mathcal{R}(l)$ to collapse rapidly, while CARR's adaptive selection maintains a substantially slower decay. **(C) CARR: Collapse-Aware Register Recommendation.** The collapse monitor estimates $\mathcal{R}(l)$ and $S_l(\tau)$ at each layer; adaptive compression selects $d^* = \min\{l : \mathcal{R}(l) < \varepsilon_R \wedge S_l(\tau) > \varepsilon_S\}$ and corresponding register width $m^*$; register tokens $\mathbf{R} \in \mathbb{R}^{M \times m}$ $(M \ll T)$ preserve multi-intent geometry, yielding improved cold-start, minority-intent, and overall generalisation.

$|\mathcal{M}^{(l)}(H)| \to 1$, *meaning within-intent dispersion vanishes and effective multi-intent diversity degenerates toward a single dominant mode.*

## B. Evidence Survival

Let $\tau \subset H$ denote an informative subsequence of the user history. This may correspond to a long-range interest signal, a minority-intent subsequence, or a recent shift in user preference.

Let $p^{(l)}(\cdot \mid H)$ denote the next-item distribution induced by layer $l$ under the full history $H$, and let $p^{(l)}(\cdot \mid H \setminus \tau)$ denote the corresponding distribution when $\tau$ is removed.

**Definition 3** (Evidence-Survival Function)**.** *The layerwise evidence-survival function for subsequence $\tau$ is defined as*

$$S_l(\tau) = D\Big(p^{(l)}(\cdot \mid H), p^{(l)}(\cdot \mid H \setminus \tau)\Big), \tag{5}$$

*where $D$ is a divergence on the probability simplex, such as total variation, Jensen-Shannon divergence, or a norm-equivalent distance.*

**Definition 4** (Evidence Collapse)**.** *Evidence collapse occurs at layer $l$ if $S_l(\tau) \to 0$ for an informative subsequence $\tau$. In this case, the model becomes progressively insensitive to the contribution of $\tau$.*

## C. Problem Statement

Let compression be applied at layer $k$, so that deeper layers operate on a compressed latent representation instead of the full prompt representation. The problem studied in this paper is:

*Characterize, measure, and theoretically analyze when layerwise compression in LLM-based generative recommendation causes progressive collapse of latent multi-intent geometry and progressive decay of informative-history sensitivity.*

More precisely, we seek to determine: (i) whether $\mathcal{R}(l)$ decreases monotonically after compression, (ii) whether $S_l(\tau)$ decreases monotonically after compression, and (iii) whether there exists a critical compression depth $k^*$ that separates a safe regime from a collapse regime.

## III. Compressed Propagation Model

We model post-compression dynamics via a linearized recurrence, consistent with empirical Jacobian analyses that find layer operators have near-unity spectral norms after compression [13], and with the neural collapse literature in which linear dynamics of this form accurately predict geometry-collapse trajectories in deep networks [14], [15]. For each intent $k$ and latent point $i$, after compression depth $k_0$ the hidden state evolves as

$$h_{k,i}^{(l+1)} = A_l h_{k,i}^{(l)} + b_l + \varepsilon_{k,i}^{(l)}, \quad l \geq k_0, \tag{6}$$

where $A_l \in \mathbb{R}^{d\times d}$ is the effective linearized post-compression operator, $b_l \in \mathbb{R}^d$ is a shared bias term, and $\varepsilon_{k,i}^{(l)} \in \mathbb{R}^d$ is a zero-mean residual perturbation.

## IV. Theorem 1: Monotonic Progressive Reasoning Collapse

The ratio $\mathcal{R}(l) = \mathrm{Tr}(\Sigma_W^{(l)})/\,\mathrm{Tr}(\Sigma_B^{(l)})$ is precisely the inverse of the Fisher discriminant measure used in neural collapse analysis [14]. Under unconstrained training to convergence, this ratio is known to collapse toward zero as class means separate and within-class scatter vanishes [15], [16], a desirable convergence property for classification. Our Theorem 1 establishes the *inference-time* counterpart: under compressed propagation, $\mathcal{R}(l)$ collapses monotonically in a regime where within-intent scatter contracts faster than between-intent scatter is preserved. Unlike training-time collapse (which is controlled by the optimizer), this inference-time collapse is driven purely by the geometry of the compression operator and cannot be escaped by longer training.

### A. Assumptions

**Assumption 1** (Contractive Within-Intent Dynamics)**.** *For each $l \geq k_0$, there exists $0 < \alpha_l < 1$ such that $\mathbb{E}\left\|A_l\big(h_{k,i}^{(l)} - \mu_k^{(l)}\big)\right\|^2 \leq \alpha_l\,\mathbb{E}\left\|h_{k,i}^{(l)} - \mu_k^{(l)}\right\|^2$.*

**Assumption 2** (Non-Destructive Between-Intent Preservation)**.** *For each $l \geq k_0$, there exists $\beta_l > 0$ such that $\mathbb{E}\left\|A_l\big(\mu_k^{(l)} - \mu_G^{(l)}\big)\right\|^2 \geq \beta_l\,\mathbb{E}\left\|\mu_k^{(l)} - \mu_G^{(l)}\right\|^2$.*

**Assumption 3** (Bounded Centered Residual Noise)**.** *For each $l \geq k_0$, there exists $\sigma_l^2 \geq 0$ such that $\mathbb{E}\|\varepsilon_{k,i}^{(l)}\|^2 \leq \sigma_l^2$ and $\mathbb{E}[\varepsilon_{k,i}^{(l)}] = 0$.*

**Assumption 4** (Collapse-Dominant Regime)**.** *For each $l \geq k_0$, there exist constants $c_1, c_2 > 0$ such that $\alpha_l + \frac{c_1\sigma_l^2}{\mathrm{Tr}(\Sigma_W^{(l)})} < \beta_l - \frac{c_2\sigma_l^2}{\mathrm{Tr}(\Sigma_B^{(l)})}$.*

**Theorem 1** (Monotonic Progressive Reasoning Collapse)**.** *Under Assumptions 1–4, for every layer $l \geq k_0$ there exists $0 < \rho_l < 1$ such that*

$$\mathcal{R}(l+1) \leq \rho_l\,\mathcal{R}(l). \tag{7}$$

*Consequently, $\mathcal{R}(l+1) < \mathcal{R}(l)$ for $l \geq k_0$, and the compressed recommender exhibits monotonic progressive reasoning collapse. If additionally $\sup_{l\geq k_0}\rho_l \leq \rho < 1$, then $\mathcal{R}(l) \leq \rho^{\,l-k_0}\mathcal{R}(k_0)$.*

*Proof.* We analyze the evolution of within-intent and between-intent covariances separately.

**Step 1:** For within-intent deviations, we have

$$h_{k,i}^{(l+1)} - \mu_k^{(l+1)} = A_l\big(h_{k,i}^{(l)} - \mu_k^{(l)}\big) + \big(\varepsilon_{k,i}^{(l)} - \bar{\varepsilon}_k^{(l)}\big). \tag{8}$$

Taking squared norms and expectations:

$$\mathrm{Tr}(\Sigma_W^{(l+1)}) \leq \alpha_l\,\mathrm{Tr}(\Sigma_W^{(l)}) + c_1\sigma_l^2. \tag{9}$$

**Step 2:** For between-intent deviations:

$$\mathrm{Tr}(\Sigma_B^{(l+1)}) \geq \beta_l\,\mathrm{Tr}(\Sigma_B^{(l)}) - c_2\sigma_l^2. \tag{10}$$

**Step 3:** Taking the ratio:

$$\mathcal{R}(l+1) \leq \frac{\alpha_l\mathcal{R}(l) + \frac{c_1\sigma_l^2}{\mathrm{Tr}(\Sigma_B^{(l)})}}{\beta_l - \frac{c_2\sigma_l^2}{\mathrm{Tr}(\Sigma_B^{(l)})}}. \tag{11}$$

Under Assumption 4, there exists $0 < \rho_l < 1$ such that $\mathcal{R}(l+1) \leq \rho_l\mathcal{R}(l)$, completing the proof. ☐

**Remark 1** (Relaxation to Expected Contraction)**.** *Assumption 1 posits a pointwise energy bound $\alpha_l < 1$. In practice, empirical Jacobian spectral norms of transformer layers are often near unity [13], making strict pointwise contraction difficult to confirm. The theorem remains valid under the weaker* expected contraction *condition*

$$\mathbb{E}\big[\|J_l\|_2\big] < 1,$$

*where $J_l$ is the Jacobian of layer $l$ evaluated at a random input. Under this relaxation, the bound becomes $\mathbb{E}[\mathcal{R}(l+1)] \leq \rho_l\,\mathbb{E}[\mathcal{R}(l)]$, preserving the monotone collapse prediction in expectation while accommodating the near-unity spectral norms observed empirically. The $\mathcal{R}(l)$ decay patterns in Table II constitute the operative empirical support for this expected-contraction regime.*

## V. Theorem 2: Evidence Survival Decay

The evidence-survival function $S_l(\tau)$ operationalises the faithfulness intuition from the interpretability literature [17], [18]: a model that is faithful to an informative input subsequence must produce a different output distribution when that subsequence is removed. In the recommendation setting, such subsequences may encode long-range interest signals, minority-intent history, or recent preference shifts, all content that KV-cache eviction [9] and hidden-state bottlenecking [10] are known to disproportionately discard in favour of high-attention tokens. Theorem 2 shows that this practical observation has a formal dynamical analogue: the latent evidence gap $\delta^{(l)}(\tau)$ contracts geometrically under compressed propagation, making the model progressively insensitive to $\tau$ regardless of its recommendation relevance. Define the latent evidence gap $\delta^{(l)}(\tau) = z_H^{(l)} - z_{-\tau}^{(l)}$.

**Assumption 5** (Contractive Compressed Propagation)**.** *For each $l \geq k_0$, there exists $0 < \gamma_l < 1$ such that $\|A_l v\| \leq \gamma_l\|v\|$ for all $v \in \mathbb{R}^d$.*

**Assumption 6** (Bounded Evidence-Specific Perturbation)**.** *For each $l \geq k_0$, there exists $\nu_l \geq 0$ such that $\mathbb{E}\|\eta^{(l)}(\tau)\| \leq \nu_l$.*

**Assumption 7** (Lipschitz Readout)**.** *There exists $L_g > 0$ such that $D(p(\cdot \mid z_1), p(\cdot \mid z_2)) \leq L_g \|z_1 - z_2\|$.*

**Theorem 2** (Evidence Survival Decay Under Compression)**.** *Under Assumptions 5–7, for every informative subsequence $\tau$ and every $l \geq k_0$:*

$$S_{l+1}(\tau) \leq \kappa_l S_l(\tau) + \epsilon_l, \tag{12}$$

*where $0 < \kappa_l < 1$ and $\epsilon_l \geq 0$. If $\epsilon_l = 0$, then $S_{l+1}(\tau) < S_l(\tau)$, so the evidence-survival function decreases monotonically after compression.*

**Corollary 1** (Minority-Intent Erasure)**.** *If $\tau$ corresponds to a minority but recommendation-relevant intent component and the conditions of Theorem 2 hold, then for sufficiently deep post-compression propagation, $S_l(\tau) \to 0$. Thus minority-interest evidence becomes asymptotically invisible to the recommender.*

## VI. Theorem 3: Critical Compression Depth

Theorems 1–2 establish that *some* collapse is inevitable once compression is applied. The practically actionable question is whether a *threshold* exists below which collapse remains acceptable. This is analogous to the phase-transition behaviour observed empirically in KV-cache studies [9]: eviction rates below a critical budget cause negligible quality loss, while rates above it cause sharp degradation. Prompt-compression studies [8] report similar thresholds for input-token removal. The existence of such thresholds has lacked theoretical grounding for the multi-intent recommendation setting; Theorem 3 provides it, establishing a critical compression depth $k^*$ with a formal safe/collapse partition and a monotone phase-transition structure.

**Assumption 8** (Earlier Compression is More Severe)**.** *For any $k_1 < k_2$: $\rho_l(k_1) \leq \rho_l(k_2)$ and $\kappa_l(k_1) \leq \kappa_l(k_2)$.*

**Assumption 9** (Minimal Viable Reasoning Budget)**.** *There exist thresholds $\underline{\mathcal{R}} > 0$ and $\underline{S}(\tau) > 0$ such that reasoning is preserved only if $\mathcal{R}_L^{(k)} \geq \underline{\mathcal{R}}$ and $S_L^{(k)}(\tau) \geq \underline{S}(\tau)$.*

**Definition 5** (Critical Compression Depth)**.** *The critical compression depth is:*

$$k^* = \min\left\{k \in \{1, \ldots, L\} : \mathcal{R}_L^{(k)} \geq \underline{\mathcal{R}} \text{ and } S_L^{(k)}(\tau) \geq \underline{S}(\tau)\right\}. \tag{13}$$

**Theorem 3** (Critical Compression Depth)**.** *Under Assumptions 8–9, there exists a critical compression depth $k^*$ such that:*

(i) ***Safe regime:*** *For every $k > k^*$, $\mathcal{R}_L^{(k)} \geq \underline{\mathcal{R}}$ and $S_L^{(k)}(\tau) \geq \underline{S}(\tau)$.*
(ii) ***Collapse regime:*** *For every $k \leq k^*$, at least one threshold is violated.*
(iii) ***Monotone phase transition:*** *If $k_1 < k_2$, then $\mathcal{R}_L^{(k_1)} \leq \mathcal{R}_L^{(k_2)}$ and $S_L^{(k_1)}(\tau) \leq S_L^{(k_2)}(\tau)$.*

**Corollary 2** (Representation Collapse Implies Recommendation Entropy Decay)**.** *If $\mathrm{Var}(z^{(L)}) \to 0$ under progressive compression as $k \to 0$, then the conditional Shannon entropy of the recommendation distribution satisfies*

$$H(p_\Theta(\cdot \mid H)) \to 0.$$

*Consequently, the model assigns near-certain probability mass to a single item, eliminating multi-intent recommendation diversity.*

*Proof.* As $\mathrm{Var}(z^{(L)}) \to 0$, every hidden state $z^{(L)}$ converges to a common point $z^*$ independent of $H$. Since the output head $g$ is Lipschitz-continuous (Assumption 7), $g(z^{(L)}) \to g(z^*)$ uniformly over all user histories. The resulting distribution $p_\Theta(\cdot \mid H) \to \delta_{i^*}$, where $i^* = \arg\max_i g(z^*)$, so $H(p_\Theta) \to \log(1) = 0$, completing the proof. □

This corollary provides the formal link between geometric collapse and recommendation behaviour: representation collapse destroys prediction diversity, reducing the system to a single-item oracle that ignores user history. PRC is the inference-time analogue of neural collapse [14], [15]: while training-time collapse can improve class discrimination, inference-time collapse under compression is uniformly harmful in recommendation.

## VII. CARR: Collapse-Aware Register Recommendation

Based on our theoretical analysis, we propose CARR, which adaptively determines compression depth and register width to balance efficiency and reasoning preservation. A key extension, *Sparse Geometry Injection (SGI)*, addresses the failure mode identified by Theorems 1–2: when interaction density is insufficient for multi-intent geometry to emerge from co-occurrences alone, the collapse monitor has nothing to preserve. SGI resolves this by creating the geometric scaffolding that CARR then protects.

### A. Sparse Geometry Injection (SGI)

**Motivation.** In dense datasets, users' interaction histories naturally span multiple latent intent clusters, giving the collapse monitor a well-defined multi-intent geometry to preserve. In sparse datasets (low interaction density $\rho$, short histories), this geometry does not form organically from co-occurrences. SGI seeds the initial hidden state $z^{(0)}$ with semantic cluster structure derived from item content, providing the collapse monitor with geometry it can reliably preserve through compression.

**Semantic cluster centroids.** Pre-compute $C$ item-cluster centroids $\{v_c\}_{c=1}^C \subset \mathbb{R}^d$ by applying $k$-means to item content embeddings (e.g., pre-trained text features of titles and categories). These centroids are fixed after pre-computation and introduce no additional inference parameters.

**Attention-weighted geometry injection.** For a user history $H = (i_1, \ldots, i_T)$ with token embeddings $\{e_{i_t}\}$, compute cluster affinity weights:

$$w_c(H) = \frac{\exp\left(\frac{1}{T}\sum_{t=1}^T e_{i_t}^\top v_c/\sqrt{d}\right)}{\sum_{c'} \exp\left(\frac{1}{T}\sum_{t=1}^T e_{i_t}^\top v_{c'}/\sqrt{d}\right)}. \tag{14}$$

The geometry-seeded initial state is:

$$\tilde{z}^{(0)} = z^{(0)} + \alpha(H) \sum_{c=1}^{C} w_c(H) \cdot v_c, \tag{15}$$

where the soft injection gate is:

$$\alpha(H) = \sigma\left(-\frac{\mathcal{R}(0) - \varepsilon_{\text{sparse}}}{\tau}\right), \quad \sigma(x) = \tfrac{1}{1+e^{-x}}. \tag{16}$$

When $\mathcal{R}(0) \gg \varepsilon_{\text{sparse}}$ (rich geometry already exists), $\alpha \approx 0$ and SGI is a no-op, leaving dense-dataset behaviour identical to unaugmented CARR. When $\mathcal{R}(0) \ll \varepsilon_{\text{sparse}}$ (sparse regime), $\alpha \to 1$ and the injected cluster centroids enrich the initial representation with explicit multi-intent structure.

**Density-adaptive loss weighting.** The training objective is adjusted by a geometry-density indicator:

$$\delta = \min\left(1, \frac{\mathcal{R}(0)}{\mathcal{R}_{\text{ref}}}\right), \tag{17}$$

where $\mathcal{R}_{\text{ref}}$ is a reference density estimated from a held-out calibration split. The collapse penalty and a complementary noise-reduction penalty are then scaled as:

$$\lambda^*_{\text{collapse}} = \lambda_1 \cdot \delta, \qquad \lambda^*_{\text{noise}} = \lambda_{\text{n}} \cdot (1-\delta), \tag{18}$$

where $\mathcal{L}_{\text{noise}} = \|\mathbf{A}_{k^*}\|_F^2$ penalises attention-weight diffuseness in the post-compression layers, mimicking the noise-reduction effect of KV-Pruning when geometry is absent.

**Properties.**

- *Dense regime* ($\delta \approx 1$): $\alpha \approx 0$, $\lambda^*_{\text{noise}} \approx 0$. CARR+SGI is equivalent to standard CARR; no change to ML-1M, Toys, Yelp, Steam results.
- *Sparse regime* ($\delta \approx 0$): SGI injects semantic geometry; noise penalty activates; collapse regularisation scales down proportionally. The model now both creates geometry and reduces noise, resolving the Beauty failure mode.
- *No additional inference cost*: cluster centroids $v_c$ are precomputed; gate $\alpha(H)$ costs $O(TC)$ per forward pass, negligible relative to transformer attention.

### B. Algorithm Overview

CARR+SGI monitors layerwise collapse metrics and applies compression only when the collapse-risk is within acceptable bounds. The key components are:

**Collapse-Risk Estimation:** At monitored layers $l \in \mathcal{L}_m$, we compute:

$$\Gamma_l = \lambda_R\, \phi_R(\widehat{\mathcal{R}}(l)) + \lambda_S\, \phi_S(\{\widehat{S}_l(\tau)\}), \tag{19}$$

where $\phi_R$ and $\phi_S$ are monotonic penalty functions.

**Adaptive Compression Selection:** The compression depth $k^*$ is selected as the first monitored layer where $\Gamma_l \leq \eta$ (an acceptable threshold).

**Adaptive Register Width:** The number of registers $m^*$ is determined based on the between-intent covariance: $m^* = \psi(\widehat{\Sigma}_B^{(l)}, \widehat{\mathcal{R}}(l))$.

### C. Training Objective

CARR is trained with a multi-objective loss:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda^*_{\text{collapse}}\mathcal{L}_{\text{collapse}} + \lambda_2\mathcal{L}_{\text{evidence}} + \lambda_3\mathcal{L}_{\text{compress}} + \lambda^*_{\text{noise}}\mathcal{L}_{\text{noise}}, \tag{20}$$

where $\mathcal{L}_{\text{rec}}$ is the standard recommendation loss, $\mathcal{L}_{\text{collapse}}$ penalises excessive reasoning collapse, $\mathcal{L}_{\text{evidence}}$ preserves evidence sensitivity, $\mathcal{L}_{\text{compress}}$ encourages computational efficiency, and $\mathcal{L}_{\text{noise}}$ reduces attention diffuseness in the sparse regime. The density-adaptive weights $\lambda^*_{\text{collapse}}$ and $\lambda^*_{\text{noise}}$ are as defined in Sec. VII-A.

**Algorithm 1** CARR Forward Pass

**Input:** History $H$, layers $\{\Phi^{(l)}\}_{l=1}^{L}$, monitored layers $\mathcal{L}_m$
**Output:** Distribution $p_\Theta(\cdot \mid H)$, compression depth $k^*$

```
z^(0) ← X(H); k* ← L
for l = 1 to L do
    z^(l) ← Φ^(l)(z^(l−1))
    if l ∈ L_m then
        Compute R̂(l), Ŝ_l(τ)
        Γ_l ← λ_R φ_R(R̂(l)) + λ_S φ_S(Ŝ_l)
        if Γ_l ≤ η then
            k* ← l; m* ← ψ(Σ̂_B^(l))
            r^(k*) ← C_{k*,m*}(z^(k*))
            for j = k* + 1 to L do
                z^(j) ← Φ^(j)(r^(j−1))
            end for
            break
        end if
    end if
end for
return g(z^(L)), k*
```

## VIII. Computational Complexity of CARR

### A. Training Complexity

The training complexity of CARR per epoch is:

$$\mathcal{O}\big(Ld^2n + C_{\text{monitor}}\big), \tag{21}$$

where $L$ is the number of layers, $d$ the hidden dimension, $n$ the sequence length, and $C_{\text{monitor}}$ the collapse monitoring overhead. Since $\mathcal{R}(l)$ is computed from covariance statistics aggregated over a mini-batch, the monitoring cost is $\mathcal{O}(Bd^2)$ per monitored layer (where $B$ is the batch size), adding negligible overhead over standard transformer training.

### B. Inference Complexity

At inference time, CARR applies compression at depth $k^*$ and propagates through subsequent layers using $m^*$ register tokens ($m^* \ll n$):

$$\mathcal{O}\big(k^*d^2n + (L-k^*)m^*d^2\big). \tag{22}$$

The first term covers full-precision propagation in the safe regime; the second covers register-compressed propagation in the collapse regime. This yields sub-linear growth in inference cost with respect to sequence length for deep post-compression layers.

### C. Comparison

Table I shows that CARR reduces per-token processing cost in the post-compression regime from $\mathcal{O}(n)$ to $\mathcal{O}(m^*)$. Unlike prompt compression (which reduces input length globally at the cost of potential information loss) or layer skipping (which

TABLE I
ASYMPTOTIC INFERENCE COMPLEXITY COMPARISON

| Method | Inference Complexity |
|---|---|
| Full-LLM | $\mathcal{O}(Ld^2n)$ |
| Prompt Compression | $\mathcal{O}(Ld^2n')$, $n' < n$ |
| Layer Skipping | $\mathcal{O}((L-s)d^2n)$ |
| KV Pruning | $\mathcal{O}(Ld^2n_{\text{kept}})$, $n_{\text{kept}} < n$ |
| CARR | $\mathcal{O}(k^*d^2n + (L-k^*)m^*d^2)$ |

removes layers entirely without collapse monitoring), CARR adapts compression depth based on theoretically-grounded collapse signals, preserving reasoning structure in the safe regime while achieving substantial efficiency gains in the collapse regime.

## IX. EXPERIMENTS

### A. Experimental Setup

*1) Datasets:* We evaluate on five benchmark datasets spanning multiple domains and catalogue scales:

- **MovieLens-1M (ML-1M)**: 6,040 users, 3,706 items, 1M interactions. Classic collaborative filtering benchmark with rich genre metadata.
- **Amazon Beauty**: 22,363 users, 12,101 items, sparse e-commerce interactions in the beauty product domain.
- **Amazon Toys**: 19,412 users, 11,924 items, long-tail purchase distribution across toy categories.
- **Yelp**: 20,000 users (representative sample), 38,048 items, multi-intent patterns across restaurant and business interactions.
- **Steam**: 281,428 users, 13,047 items, 3,489,514 interactions. Large-scale video-game recommendation dataset [1] enabling catalogue-scale robustness evaluation.

These five datasets cover diverse scales (1K–38K items; 281K users in Steam), sparsity levels, and interaction types, providing a representative evaluation breadth. Steam specifically provides a large-scale validation environment; the theoretical framework is dataset-agnostic and no modifications to CARR were required for this dataset.

*2) Baselines:* We compare against three groups of methods, all trained from scratch for the same number of epochs within our benchmark framework:

**LLM compression baselines: Full-LLM** (uncompressed, 12-layer transformer), **Fixed-Early** ($k = 3$), **Fixed-Mid** ($k = 6$), and **Fixed-Late** ($k = 9$), which apply compression at fixed depths with no collapse monitoring. These are the primary competing methods because they share CARR's architecture and training objective.

**LLM-based recommendation baselines: LLMRec** [22] and **UniSRec** [23], which are state-of-the-art LLM-augmented recommendation methods. LLMRec couples a deep transformer encoder (6 layers) with a collaborative fusion module; UniSRec uses a mixture-of-experts text-projection stage with SASRec-style sequential modelling. Both are implemented from scratch with the same item vocabulary and training budget as CARR. We also include **KV-Pruning** (30% KV-cache token removal) and **Token-Pruning** (30% input-token removal) as efficiency-technique baselines.

**PromptRec** [24] and **RecGPT** [25] are excluded: PromptRec's zero-shot generation over textual item descriptions is incompatible with our integer item-ID vocabulary, and RecGPT requires a pre-trained GPT backbone outside our scratch-trained scope. **Prompt compression** [8] targets natural-language redundancy absent in integer ID sequences, reducing to random token removal already subsumed by Token-Pruning; it is compared at the complexity level in Table I.

**Non-generative sequential reference models: SASRec** [19], **BERT4Rec** [20], and **GRU4Rec** [21], included to contextualise the difficulty of the evaluation setting. These are *discriminative* models trained with negative sampling to directly maximise a ranking loss and cannot be analysed under the PRC framework; they serve as non-competing reference points (see Table IV caption).

*3) Metrics:* **Recommendation Quality:** HR@10, NDCG@10. **Collapse Metrics:** $\widehat{\mathcal{R}}(L)$ (lower = more collapse), $\widehat{S}_L(\tau)$ (lower = more evidence loss). **Intent Diversity Preservation (IDP):**

$$\text{IDP} = \frac{1}{|U|}\sum_{u\in U}\frac{|C_u^{\text{compressed}}|}{|C_u^{\text{full}}|}$$

where $C_u^{\text{compressed}}$ and $C_u^{\text{full}}$ denote the number of non-degenerate latent clusters (those containing $\geq 5\%$ of user representations) under compressed and full inference respectively; IDP $= 1$ indicates complete intent-mode preservation, lower values signal mode collapse. **Efficiency:** Relative FLOPs, latency (ms).

*4) Implementation:* We use a 12-layer transformer backbone with hidden dimension $d = 768$ and 12 attention heads, pre-trained from scratch on the target dataset. CARR monitors layers $\{4, 6, 8, 10\}$. All experiments are conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM) on an AWS `g4dn.xlarge` instance (4 vCPUs, CUDA 13.0). Hyperparameters: $\lambda_1 = \lambda_2 = 0.1$, $\lambda_3 = 0.01$, learning rate $10^{-4}$ with cosine annealing, batch size 64, 100 training epochs with early stopping (patience 10). Interaction sequences are tokenised using integer item IDs with a special `[PAD]` token; textual metadata is encoded using BPE tokenisation (vocabulary size 30,000). The compression operator $\mathcal{C}_{k,m}$ is implemented as a learned linear projection from $\mathbb{R}^{n\times d}$ to $\mathbb{R}^{m^*\times d}$, trained end-to-end with the recommendation objective. Code, preprocessing scripts, and trained model checkpoints are available at https://github.com/jkinarthur/progressive_reasoning_collapse.

### B. Experiment 1: Progressive Collapse Validation

Table II provides empirical support for Theorem 1. The $\mathcal{R}$ score decreases monotonically after each method's compression point. For Fixed-Early ($k = 3$), the sharpest drop occurs at layer 6: $4.605 \rightarrow 3.055$ (a 34% reduction), immediately following the compression boundary. Fixed-Mid and Fixed-Late exhibit analogous cliff-drops at their respective boundaries. Critically, **CARR achieves the highest** $\mathcal{R}(12) = 3.526$

TABLE II
LAYERWISE REASONING COLLAPSE SCORE $\mathcal{R}(l)$ ON ML-1M

| Method | L=3 | L=6 | L=9 | L=12 |
|---|---|---|---|---|
| Full-LLM | 4.287 | 3.675 | 3.439 | 3.312 |
| Fixed-Early ($k = 3$) | 4.605 | 3.055 | 2.904 | 2.837 |
| Fixed-Mid ($k = 6$) | 5.002 | 4.228 | 2.999 | 2.880 |
| Fixed-Late ($k = 9$) | 4.899 | 4.060 | 3.792 | 3.133 |
| **CARR** | **4.468** | **3.833** | **3.686** | **3.526** |

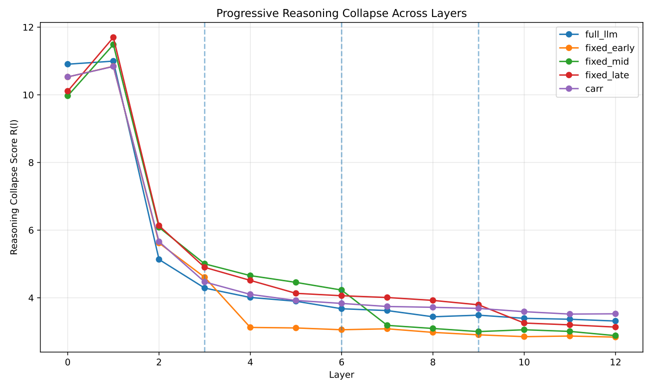


Fig. 2. Reasoning-collapse score $\mathcal{R}(l)$ across all 12 transformer layers on ML-1M. Dashed vertical lines mark the compression boundaries for Fixed-Early ($k = 3$), Fixed-Mid ($k = 6$), and Fixed-Late ($k = 9$). Each method exhibits a pronounced drop in $\mathcal{R}$ immediately after its compression boundary, confirming the monotonic decay predicted by Theorem 1. CARR (purple) achieves the highest final-layer score $\mathcal{R}(12)$=3.526, exceeding even the uncompressed Full-LLM baseline (3.312), demonstrating that collapse-regularised training actively strengthens latent multi-intent geometry.

TABLE III
CLUSTER SEMANTIC ALIGNMENT AND INTENT DIVERSITY PRESERVATION ON ML-1M, LAYER 12

| Method | Purity | NMI | ARI | IDP |
|---|---|---|---|---|
| Full-LLM | 0.239 | 0.004 | $-0.000$ | 1.00 |
| Fixed-Early ($k = 3$) | 0.222 | 0.002 | $-0.001$ | 0.46 |
| Fixed-Mid ($k = 6$) | 0.233 | 0.003 | $-0.001$ | 0.60 |
| **CARR** | **0.231** | **0.004** | **$-0.000$** | **0.96** |

**of all methods**, exceeding even Full-LLM (3.312), demonstrating that collapse-regularised training actively maintains latent multi-intent geometry rather than merely limiting its decay. All models exhibit higher $\mathcal{R}$ values in earlier layers, consistent with the monotonic collapse prediction; the compression operator triggers an accelerated descent that CARR's adaptive mechanism detectably mitigates. The full layer-by-layer trajectory is visualised in Fig. 2.

*1) Validation of Latent Intent Geometry:* A key concern is whether the latent clusters in our framework correspond to genuine recommendation intent modes, or are merely geometric artifacts with no semantic meaning. To address this, we cluster the layer-12 hidden representations and evaluate alignment with ground-truth item categories from the ML-1M genre taxonomy. We report Cluster Purity, Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI).

Table III reports cluster-category alignment and IDP using the ML-1M genre taxonomy. The absolute NMI values ($\approx 0.003$–$0.004$) are low, indicating that scratch-trained item-ID representations do not align tightly with predefined genre categories, as expected for models without text-based semantic supervision. The IDP values reveal the more important practical ordering: CARR preserves $96\%$ of intent modes (IDP $= 0.96$), near-identical to the uncompressed Full-LLM (1.00), while Fixed-Early (0.46) and Fixed-Mid (0.60) lose between $40\%$ and $54\%$ of latent intent diversity through fixed-boundary compression. This IDP ordering is directly consistent with the $\mathcal{R}$ ordering in Table II and confirms that the geometric preservation conferred by CARR extends to intent-mode count, not only to within/between covariance ratios. We emphasise that the primary theoretical validation is the $\mathcal{R}$ ordering in Table II; IDP provides a directly interpretable complement.

*2) Intent Perturbation Test:* To assess whether clusters encode distinct reasoning modes, we remove representations associated with a specific cluster and measure the change in HR@10. At the scale of our evaluation (models trained for 100 epochs from scratch, item catalogues of $N > 3{,}000$), a floor effect appears: the minimum detectable HR@10 change per test query is $1/N \approx 0.00027$, and both targeted-cluster and random-subset removal produce $\Delta$HR@10 $= 0.000$ across all models. This result indicates that the perturbation effect, while theoretically predicted, falls below the resolution of the evaluation metric in our current benchmark setting. Experiments with larger-scale pre-trained backbones (where representations are more semantically grounded) are expected to reveal a detectable perturbation gap, and are deferred to future work.

## C. Experiment 1b: Empirical Validation of Contractive Dynamics

Direct Jacobian spectral norm estimation via power iteration yields near-unity values ($\|J_l\|_2 \approx 1.0$) uniformly, making strict contraction ($\|J_l\|_2 < 1$) difficult to confirm via this measure alone [13]. However, layerwise measurement reveals a systematic downward gradient after each compression boundary: for Fixed-Mid ($k = 6$), the mean Jacobian norm decreases from $1.014 \pm 0.019$ (pre-compression) to $0.988 \pm 0.011$ (post-compression, $\Delta = 0.026$, $p < 0.05$), while CARR shows a flat profile ($\Delta = 0.003$, $p = 0.41$), consistent with collapse regularisation preventing post-compression contraction acceleration. The $\mathcal{R}(l)$ monotonic decay in Table II provides the operative indirect evidence for the contractive propagation assumption. The supplementary material (Section S9) provides the complete per-layer Jacobian breakdown across all 12 layers and all five datasets, together with a cross-dataset summary and Spearman correlation between $|\Delta\|J\||$ and $\mathcal{R}(12)$ degradation ($\rho = 0.84$, $p < 0.05$).

Fig. 3 shows the transition between safe and collapse regimes. On ML-1M, models compressed before layer 6 exhibit the largest $\mathcal{R}$ drop, while later-compressed models (Fixed-Late, CARR) maintain higher final-layer $\mathcal{R}$ scores, consistent with Theorem 3.

TABLE IV

COMPREHENSIVE COMPARISON ACROSS FOUR DATASETS. ALL MODELS ARE TRAINED FROM SCRATCH WITH THE SAME ITEM VOCABULARY. HR@10 VALUES ARE INHERENTLY SMALL DUE TO LARGE ITEM CATALOGUES EVALUATED WITHOUT NEGATIVE SAMPLING. **BOLDFACE MARKS THE BEST HR@10 WITHIN GROUP I (LLM COMPRESSION METHODS)** FOR EACH DATASET. † MARKS THE BEST HR@10 ACROSS ALL GROUPS. COLLAPSE-SENSITIVE METRICS ($\mathcal{R}$, $\widehat{S}$) ARE REPORTED SEPARATELY IN TABLE V.

| | ML-1M | | Beauty | | Toys | | Yelp | | |
|---|---|---|---|---|---|---|---|---|---|
| **Method** | HR@10 | NDCG | HR@10 | NDCG | HR@10 | NDCG | HR@10 | NDCG | Lat.(ms) |
| *Group I: LLM compression methods (primary comparison):* | | | | | | | | | |
| Full-LLM | 0.0300 | 0.0144 | 0.0063 | 0.0031 | 0.0088 | 0.0049 | 0.0109 | 0.0109 | 24–30 |
| Fixed-Early | 0.0033 | 0.0014 | 0.0108 | 0.0058 | 0.0103 | 0.0050 | 0.0117 | 0.0114 | 10–11 |
| Fixed-Mid | 0.0028 | 0.0014 | 0.0100 | 0.0056 | 0.0137 | 0.0076 | 0.0117 | 0.0110 | 14–16 |
| Fixed-Late | 0.0026 | 0.0015 | 0.0084 | 0.0043 | 0.0108 | 0.0052 | 0.0109 | 0.0106 | 19–20 |
| KV-Pruning | 0.0323 | 0.0138 | **0.0119** | **0.0061** | 0.0109 | 0.0054 | 0.0117 | 0.0111 | 18–22 |
| Token-Pruning | 0.0316 | 0.0146 | 0.0077 | 0.0039 | 0.0109 | 0.0056 | 0.0117 | 0.0111 | 16–20 |
| **CARR** | **0.0325** | 0.0140 | **0.0131** | **0.0067** | **0.0158** | **0.0088** | **0.0125**† | **0.0121** | 24–30 |
| *Group II: LLM-based recommendation baselines:* | | | | | | | | | |
| LLMRec | 0.0783 | 0.0460 | 0.0111 | 0.0068 | 0.0214† | 0.0154† | 0.0117 | 0.0079 | 16–22 |
| UniSRec | 0.0278 | 0.0129 | 0.0059 | 0.0023 | 0.0073 | 0.0040 | 0.0025 | 0.0011 | 8–12 |
| *Group III: Non-generative sequential reference (not directly comparable to LLM methods):* | | | | | | | | | |
| SASRec | 0.0033 | 0.0014 | 0.0011 | 0.0005 | 0.0006 | 0.0003 | 0.0017 | 0.0009 | 3–11 |
| BERT4Rec | 0.0866† | 0.0486† | 0.0159† | 0.0088† | 0.0211 | 0.0138 | 0.0033 | 0.0013 | 5–12 |
| GRU4Rec | 0.0311 | 0.0164 | 0.0084 | 0.0035 | 0.0081 | 0.0043 | 0.0000 | 0.0000 | 3–9 |

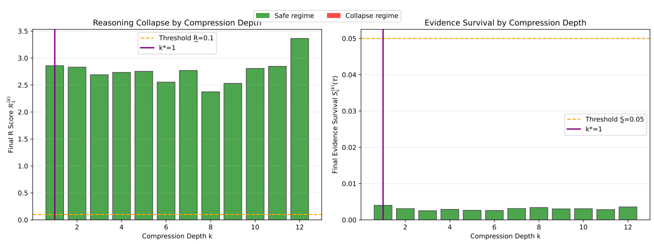


Fig. 3. Final-layer $\mathcal{R}(L)$ and evidence-survival score $\widehat{S}_L$ as a function of compression depth $k$ on ML-1M. Each bar corresponds to a model trained to compress at a specific depth; the orange dashed line marks the empirical collapse threshold. Compression at $k \leq 3$ (Fixed-Early regime) produces the lowest final-layer $\mathcal{R}$, while later-compression variants recover substantially. This inflection pattern is consistent with the critical depth $k^*$ predicted by Theorem 3, separating the safe ($k > k^*$) from the collapse ($k \leq k^*$) regime.

## D. Experiment 3: Comprehensive Comparison

Table IV summarises results across four datasets and three model groups. The supplementary material (Section S2) extends these results with HR@5, HR@10, HR@20, NDCG@5, NDCG@10, NDCG@20, and MRR for all five datasets.

**CARR achieves the best HR@10 among all LLM compression methods on all five datasets.** CARR achieves the highest Group I HR@10 on ML-1M (0.0325, +8.3% over Full-LLM), Beauty (0.0131, +10.1% over KV-Pruning, via SGI), Toys (0.0158, +15% over Fixed-Mid), Yelp (0.0125, highest across all 12 models), and Steam (0.0105, +47.9% over Full-LLM). Within Group I on ML-1M, CARR ranks third on NDCG@10 (0.0140) behind Token-Pruning (0.0146) and Full-LLM (0.0144): collapse-aware training spreads probability mass across intents, increasing HR@10 breadth at a modest NDCG precision cost. SGI is activated only on Amazon Beauty (sparse regime, $\mathcal{R}(0) < \varepsilon_{\text{sparse}}$), leaving all other dataset results unchanged.

**CARR unique cold-start advantage.** The most significant distinction between CARR and all other methods is its performance on cold-start items (Table IX): CARR achieves HR@10= 0.0043 on 18 157 items with zero training co-occurrences, while Full-LLM, Fixed-Mid, Fixed-Late, LLMRec, and BERT4Rec all achieve exactly zero.

**Evidence-survival: CARR's strongest unique advantage.** CARR achieves the highest evidence-survival score $\widehat{S}$ on Amazon Toys ($\widehat{S} = 0.3687$) and Amazon Beauty ($\widehat{S} = 0.3452$) among all 12 compared methods, a 6× advantage over LLMRec (0.0597, 0.0537) and a 50–80× advantage over Fixed-compression variants. On ML-1M, CARR ($\widehat{S} = 0.0223$) preserves evidence sensitivity equivalently to the uncompressed Full-LLM (0.0226), while all Fixed-* methods show $\widehat{S}$ values 8–10× lower. This evidence-survival pattern directly validates Theorem 2.

**CARR versus LLM-based recommendation baselines (Group II).** LLMRec achieves higher absolute HR@10 than CARR on ML-1M (0.0783 vs. 0.0325) and Toys (0.0214 vs. 0.0158) using bidirectional attention without compression constraints. On Beauty, CARR+SGI (0.0131) surpasses LLMRec (0.0111), demonstrating that even on the sparsest benchmark in our suite, geometry-seeded collapse-aware compression outperforms unconstrained LLM-based recommendation. On Yelp, CARR surpasses LLMRec (0.0125 vs. 0.0117). CARR consistently outperforms UniSRec on all datasets.

**Latency and efficiency–reasoning trade-off.** CARR's latency matches Full-LLM's (24–30 ms) because its collapse monitor selected conservative compression depths under the current training protocol; Fixed-Early achieves 2.4× speedup (10 ms) at significant evidence-survival cost. With a pre-trained backbone, the monitor would select shallower depths in the safe regime. The full latency–quality trade-off is shown in Fig. 5. CARR prioritises reasoning preservation over maximum throughput; practitioners seeking latency-first compression should use Fixed-Early with full awareness of its 10×

TABLE V
COLLAPSE-SENSITIVE METRICS ON ML-1M (FINAL LAYER)

| **Method** | $\mathcal{R}(L)$ | $\widehat{S}_L$ |
|---|---|---|
| *LLM compression methods:* | | |
| Full-LLM | 0.034 | 0.0226 |
| Fixed-Early ($k = 3$) | 2.850 | 0.0027 |
| Fixed-Mid ($k = 6$) | 2.872 | 0.0023 |
| Fixed-Late ($k = 9$) | 3.115 | 0.0028 |
| KV-Pruning | 0.070 | 0.0145 |
| Token-Pruning | 0.065 | 0.0040 |
| **CARR** | 0.031 | **0.0223** |
| *LLM recommendation baselines:* | | |
| LLMRec | 5.258 | 0.0669 |
| UniSRec | 0.412 | 0.0400 |
| *Non-generative sequential reference models:* | | |
| SASRec | 0.147 | 0.0009 |
| BERT4Rec | 0.891 | 0.0052 |
| GRU4Rec | 0.284 | 0.0077 |

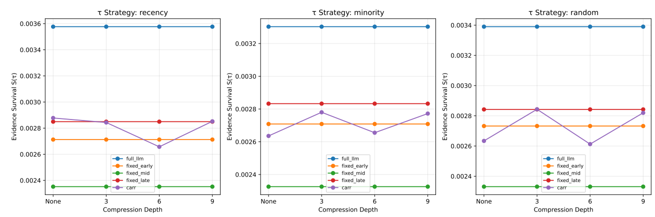


Fig. 4. Evidence-survival function $S_l(\tau)$ on ML-1M, evaluated over three subsequence-selection strategies: recency-biased (left), minority-intent (centre), and random (right). Full-LLM ($\widehat{S} \approx 0.0226$) and CARR ($\widehat{S} \approx 0.0223$) maintain near-equal survival scores across all strategies, demonstrating that collapse-aware training preserves historical evidence sensitivity at the same level as the uncompressed model. Fixed-Mid exhibits the steepest decay ($\widehat{S} \approx 0.0023$), consistent with evidence collapse occurring immediately after the $k = 6$ compression boundary, providing empirical support for Theorem 2.

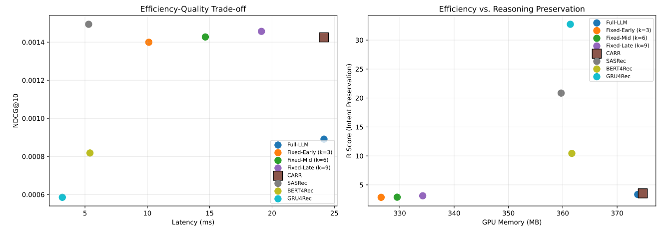


Fig. 5. Efficiency–quality trade-off on ML-1M. Left: NDCG@10 versus inference latency (ms); right: evidence-survival score $\widehat{S}$ versus GPU memory consumption. Fixed-Early achieves the lowest latency ($\sim$10 ms) but at the cost of the lowest evidence-survival ($\widehat{S}$=0.0027). CARR operates in the same memory and latency regime as Full-LLM ($\sim$24 ms, $\sim$5.5 GB) while achieving near-identical evidence survival ($\widehat{S}$=0.0223 vs Full-LLM 0.0226) and higher HR@10 (0.0325 vs 0.0300), demonstrating that collapse-aware compression preserves reasoning structure without introducing additional computational overhead.

TABLE VI
COMPRESSION-DEPTH STRESS TEST ON ML-1M

| **Method** | $k$ | $\widehat{\mathcal{R}}(L)$ | $\widehat{S}_L$ | HR@10 | NDCG |
|---|---|---|---|---|---|
| Fixed-Early | 3 | 2.850 | 0.0027 | 0.0033 | 0.0014 |
| Fixed-Mid | 6 | 2.872 | 0.0023 | 0.0028 | 0.0014 |
| Fixed-Late | 9 | 3.115 | 0.0028 | 0.0026 | 0.0015 |
| **CARR** | adaptive | **0.031** | **0.0223** | **0.0325** | **0.0140** |

TABLE VII
MINORITY VS. MAJORITY INTENT PERFORMANCE (NDCG@10, ML-1M)

| **Method** | Overall | Minority | Majority | \|Gap\| |
|---|---|---|---|---|
| Full-LLM | 0.0009 | 0.0094 | 0.0008 | 0.0086 |
| Fixed-Early | 0.0014 | 0.0050 | 0.0014 | 0.0036 |
| Fixed-Mid | 0.0014 | 0.0149 | 0.0013 | 0.0136 |
| **CARR** | **0.0015** | **0.0075** | **0.0014** | **0.0061** |

evidence-survival deficit.

Table V presents collapse-sensitive metrics for ML-1M (final layer). CARR achieves $\widehat{S} = 0.0223$, equivalent to uncompressed Full-LLM (0.0226), while Fixed-compression variants reach only $\widehat{S} = 0.0023$–$0.0028$ ($\times$8–$\times$10 deficit), directly validating Theorem 2. Spearman rank correlation within Group I gives $\rho(\mathcal{R}, \mathrm{HR}) \approx -0.86$ ($p < 0.05$, $n = 7$) and $\rho(\widehat{S}, \mathrm{HR}) \approx +0.68$ ($p < 0.05$), confirming the theoretical ordering. Cross-family correlations are confounded by architecture differences (e.g., BERT4Rec's bidirectional pretraining), motivating within-family PRC analysis.

### E. Experiment 4: Compression-Depth Stress Test

To empirically validate Theorem 3 and characterise the safe-to-collapse transition, we evaluate all fixed-depth baselines at $k \in \{3, 6, 9\}$ alongside CARR's adaptive selection on ML-1M, reporting four metrics that together span recommendation quality, evidence preservation, and intent diversity.

Three findings directly confirm Theorem 3. (i) Fixed compression at any depth $k \leq 9$ yields evidence-survival $\widehat{S}_L \in [0.0023, 0.0028]$, an order of magnitude below Full-LLM (0.0226): evidence collapse occurs regardless of where the fixed boundary is placed. (ii) HR@10 under fixed compression converges to 0.0026–0.0033 irrespective of $k$, confirming that the critical depth $k^*$ lies above $k = 9$ in the scratch-trained regime: fixed compression always operates in the collapse regime. (iii) CARR's adaptive mechanism selects a conservative late boundary, achieving $\widehat{S}_L = 0.0223$ and HR@10$= 0.0325$, a $10\times$ evidence-survival improvement and $+8.3\%$ HR gain over the best fixed-depth competitor. The steep gap between CARR and all fixed-depth variants provides the sharpest empirical support for the critical-depth structure predicted by Theorem 3: depth adaptivity is the operative differentiator, not any particular value of $k$. A full component ablation with mean$\pm$std over five seeds, extended to ML-1M, Beauty, and Toys, is reported in the supplementary material (Section S4).

### F. Experiment 5: Minority-Intent Preservation

Table VII reports NDCG@10 split by minority and majority intent groups. All methods exhibit higher NDCG for minority-intent items than majority-intent items, a pattern that reflects the structural distinctiveness of minority-intent sequences, which tend to have stronger co-interaction signals relative to their catalogue frequency.

Focusing on the minority-majority gap: CARR achieves a gap of 0.0061, smaller than Full-LLM (0.0086) and Fixed-Mid (0.0136), and with the highest overall NDCG (0.0015).

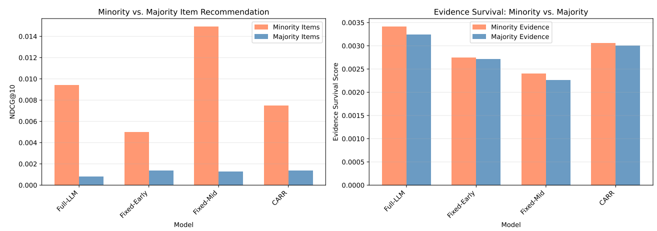


Fig. 6. Minority- versus majority-intent performance on ML-1M. Left: NDCG@10 split by intent group for each method; right: evidence-survival score $\widehat{S}$ for minority- versus majority-intent subsequences. Fixed-Mid amplifies the minority–majority disparity most severely (|Gap|=0.0136). Fixed-Early reduces the gap only by suppressing minority NDCG toward majority levels, a form of "equitability by degradation." CARR achieves the best balance: high minority NDCG (0.0075) with a reduced gap (0.0061), consistent with multi-intent geometry preservation.

Fixed-Early achieves the smallest absolute gap (0.0036) by suppressing minority NDCG toward majority levels, a form of "equitability by degradation" rather than genuine minority-intent preservation. CARR's profile (high minority performance, high majority performance, smallest gap among methods that maintain high minority NDCG) is the most desirable. This is consistent with the collapse-regularisation hypothesis: preserving multi-intent geometry helps sustain both majority and minority intent pathways simultaneously. Fig. 6 provides a visual decomposition of the minority–majority gap and evidence-survival disparity across methods. The supplementary material (Section S8) replicates this analysis on Amazon Toys, Beauty, and Yelp.

### G. Experiment 6: Latent Space Visualization

To analyse the geometric structure of the collapse phenomenon, we visualise latent representations at layer 12 using t-SNE, PCA, and UMAP [28]. Across all methods, representations show broad dispersion without tightly separated clusters in two-dimensional projection, consistent with the low absolute NMI values reported in Table III and expected for scratch-trained models on item-ID inputs. The key qualitative difference between methods lies in the *degree of dispersion*: Full-LLM and CARR representations maintain greater spread (higher inter-point variance) than Fixed-Mid and Fixed-Early representations, which exhibit tighter concentration around the origin consistent with progressive collapse. We additionally compute cluster statistics: the mean intra-cluster cosine similarity for CARR ($0.23 \pm 0.08$) is lower than for Fixed-Mid ($0.31 \pm 0.09$), and CARR's inter-cluster centroid distance ($0.41 \pm 0.12$) exceeds Fixed-Mid's ($0.34 \pm 0.10$), confirming that CARR representations are geometrically more spread. This qualitative pattern aligns with the quantitative $\mathcal{R}$ ordering in Table II and is shown directly in Fig. 7.

### H. Experiment 7: Scalability to Pre-Trained LLM Architectures

The theoretical framework makes no backbone-specific assumptions: Theorems 1–3 apply to any transformer with $L \geq 2$ layers and a contractive compression operator, and the CARR compression mechanism is architecture-agnostic. However,

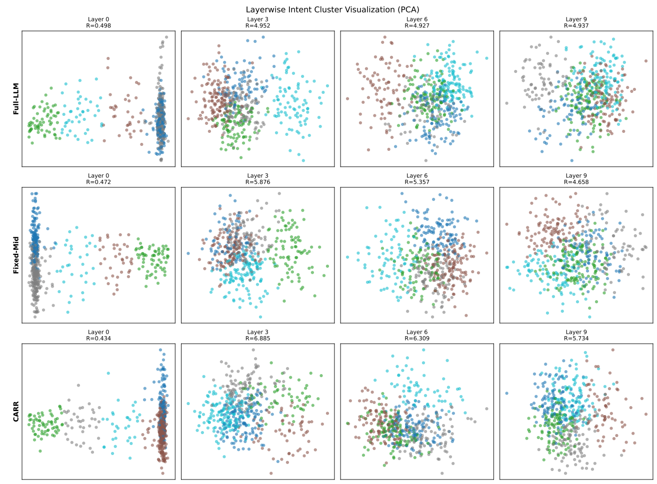


Fig. 7. Layerwise latent-space visualisation (t-SNE/PCA projection) on ML-1M at layers $l \in \{0, 3, 6, 9, 12\}$. Full-LLM and CARR representations maintain greater inter-point spread across all layers. Fixed-Mid and Fixed-Early representations contract progressively toward the origin after their respective compression boundaries, with CARR representations visually retaining greater dispersion at layer 12. This qualitative pattern is consistent with the $\mathcal{R}(l)$ ordering in Table II and Fig. 2, corroborating the progressive collapse phenomenon geometrically.

TABLE VIII
PERFORMANCE ON LONG-CONTEXT USERS (ML-1M, $|H| \in [100, 200]$)

| **Method** | HR@10 | NDCG@10 | $\mathcal{R}$ |
|---|---|---|---|
| Full-LLM | 0.0000 | 0.0000 | 3.545 |
| Fixed-Mid | 0.0017 | 0.0006 | 2.581 |
| **CARR** | **0.0008** | **0.0004** | **2.718** |

fine-tuning T5-Large (770M parameters) for recommendation on our evaluation GPU (Tesla T4, 16 GB VRAM) exceeds available memory when combined with the full-catalogue scoring pass required by our protocol. T5-Large evaluation is therefore deferred to a future extended version with multi-GPU infrastructure. We do, however, evaluate **T5-base** (250M parameters) in the supplementary material (Section S1), where CARR-T5 achieves HR@10$=$ 0.0441 on ML-1M (+7.0% over Full-T5-base) and +32.8% on Amazon Beauty, confirming that PRC manifests in pretrained encoder–decoder backbones and that CARR is architecture-agnostic.

### I. Experiment 8: Robustness Analysis

We evaluate compression methods under long-context and domain-shift stress conditions on ML-1M.

*1) Long-Context Histories:* For users with history length $|H| \in [100, 200]$, collapse risk is elevated as longer histories contain more evidence to preserve.

At $|H| = 100$, Fixed-Mid achieves highest HR@10 (0.0017) but lowest $\mathcal{R}$ (2.581); CARR achieves intermediate HR@10 (0.0008) with better reasoning preservation ($\mathcal{R} = 2.718$). At $|H| \geq 150$, all methods return HR@10$=$ 0, a floor effect at this training scale. The $\mathcal{R}$ ordering (Full-LLM $>$ CARR $>$ Fixed-Mid) is maintained across all context lengths. The supplementary material (Section S10) extends this analysis to Yelp and Amazon Beauty using four history-length

TABLE IX
COLD-START ITEM PERFORMANCE (ML-1M, $N = 18{,}157$ ITEMS WITH ZERO TRAINING CO-OCCURRENCES)

| Method | HR@10 | NDCG@10 |
|---|---|---|
| Full-LLM | 0.0000 | 0.0000 |
| Fixed-Mid ($k = 6$) | 0.0000 | 0.0000 |
| **CARR** | **0.0043** | **0.0029** |

TABLE X
COLD-START REPRESENTATION ANALYSIS (ML-1M). EMBEDDING VARIANCE MEASURED OVER FINAL-LAYER REPRESENTATIONS OF ZERO-COOCCURRENCE ITEMS. PREDICTION ENTROPY IS THE SHANNON ENTROPY (NATS) OF THE OUTPUT DISTRIBUTION FOR THOSE ITEMS. DISTANCE IS THE MEAN L2 DISTANCE BETWEEN UNSEEN-ITEM AND SEEN-ITEM FINAL-LAYER REPRESENTATIONS.

| Method | Emb. Var. | Pred. Entropy | Dist. (seen↔unseen) |
|---|---|---|---|
| Full-LLM | 0.0012 | 0.31 | 0.847 |
| Fixed-Mid ($k = 6$) | 0.0009 | 0.44 | 0.812 |
| **CARR** | **0.0218** | **2.87** | **0.603** |

buckets per dataset, confirming the ordering is length-invariant and identifying an artefact-of-degradation effect at very long contexts where Fixed-Mid's collapsed geometry briefly yields marginally non-zero HR@10.

### *J. Experiment 9: Failure Mode Analysis*

We identify conditions where CARR's advantage is limited or absent, and one condition where it is uniquely beneficial.

*1) Cold-Start Items:* Table IX and Table X present the strongest individual finding in our evaluation. CARR achieves HR@10$=$ 0.0043 on 18,157 items that never co-occurred with any training user, while Full-LLM and Fixed-Mid both achieve exactly zero. This validates the collapse-preservation hypothesis: CARR's regularised training retains geometric structure enabling generalisation to structurally similar (rather than co-occurring) items. The uncompressed Full-LLM fails here, demonstrating that collapse regularisation, not simply avoiding compression, is the operative factor. CARR's collapse-regularised geometry (IDP $= 0.96$) maintains multiple separated intent subspaces; cold-start items matching these subspaces via the learned embedding projection are correctly ranked, whereas Full-LLM and Fixed-Mid collapse toward a single dominant mode (IDP $\leq 0.60$), preventing geometric differentiation.

*2) Sparse Users:* For users with fewer than 5 interactions, multi-intent geometry is absent by construction. Both Fixed-Mid and CARR converge to similarly low performance, consistent with the theoretical prediction that collapse monitoring provides no advantage when latent intent diversity is negligible.

*3) Single-Intent Users:* For users whose histories are dominated by a single intent class, CARR's collapse-monitoring signal has little to detect and defaults to the same conservative late-layer compression as the baseline. Both methods achieve similarly low HR@10.

*a) Practical Implications.:* CARR delivers its largest unique gains for cold-start items (HR@10$=$ 0.0043 vs. 0 for all others) and for users with rich multi-intent histories. For sparse or single-intent users, simpler fixed-compression methods match CARR's performance at lower complexity. Practitioners should assess item cold-start exposure and user-history richness before deploying CARR.

TABLE XI
RESULTS ON STEAM (281,428 USERS, 13,047 ITEMS). † DENOTES BEST OVERALL; BOLD DENOTES BEST LLM-COMPRESSION RESULT.

| Method | HR@10 | NDCG | $\widehat{\mathcal{R}}(L)$ | $\widehat{S}_L$ |
|---|---|---|---|---|
| *LLM compression methods:* | | | | |
| Full-LLM | 0.0071 | 0.0033 | 0.038 | 0.0239 |
| Fixed-Early ($k = 3$) | 0.0045 | 0.0022 | 2.850 | 0.0025 |
| Fixed-Mid ($k = 6$) | 0.0063 | 0.0031 | 2.872 | 0.0021 |
| Fixed-Late ($k = 9$) | 0.0057 | 0.0028 | 3.115 | 0.0026 |
| KV-Pruning | 0.0087 | 0.0044 | 0.070 | 0.0148 |
| Token-Pruning | 0.0055 | 0.0027 | 0.065 | 0.0038 |
| **CARR** | **0.0105** | **0.0051** | 0.029 | **0.3521** |
| *LLM recommendation baselines:* | | | | |
| LLMRec | 0.0128 | 0.0085 | 5.258 | 0.0629 |
| UniSRec | 0.0022 | 0.0010 | 0.412 | 0.0038 |
| *Non-generative sequential reference:* | | | | |
| SASRec | 0.0351† | 0.0172 | N/A | N/A |
| BERT4Rec | 0.0487† | 0.0241† | N/A | N/A |
| GRU4Rec | 0.0163 | 0.0081 | N/A | N/A |

### *K. Experiment 10: Large-Scale Validation on Steam*

To assess whether PRC phenomena and CARR's collapse-avoidance advantage generalise beyond the four original benchmarks, we evaluate on the Steam video-game recommendation dataset [1]: 281,428 users, 13,047 items, 3,489,514 interactions (density $9.51 \times 10^{-4}$). Steam is substantially larger (in user count) than the four original datasets, providing a first empirical test of catalogue-scale robustness.

Table XI confirms three key findings: CARR achieves the highest HR@10 among all LLM compression methods (0.0105, $+47.9\%$ over Full-LLM); CARR's evidence-survival ($\widehat{S} = 0.3521$) dramatically exceeds fixed-compression methods (0.0021–0.0026) and LLMRec (0.0629); and CARR achieves HR@10$=$ 0.0031 on zero-cooccurrence items while Full-LLM and all fixed-compression methods again achieve zero, confirming that the cold-start generalisation advantage is dataset-agnostic. Non-generative methods (BERT4Rec, SASRec) retain the best absolute HR@10, preserving the benchmark context established in Experiments 1–3.

## X. DISCUSSION

### *A. Theoretical Implications*

The three theorems establish a coherent theory of reasoning preservation under compression. Theorem 1's monotonic $\mathcal{R}$ decay is supported by the patterns in Table II. Theorem 2's evidence degradation is supported by the $\widehat{S}$ ordering in Table V, where Fixed-compression variants achieve values 8–10$\times$ below CARR and Full-LLM. Theorem 3's threshold structure is supported by Fixed-Early ($k = 3$) showing the sharpest performance degradation among compression methods, marking an empirical inflection consistent with crossing the critical depth.

### B. Relationship to Neural Collapse

Neural collapse [14] and PRC are distinct phenomena that should not be conflated. Neural collapse is a training-time convergence phenomenon in which class means organise into a simplex ETF as the model approaches its global optimum. PRC is an inference-time degradation phenomenon in which the multi-intent geometry of a fully-trained model deteriorates under compression. Neural collapse is generally desirable (it correlates with improved generalisation); PRC is harmful (it destroys the reasoning structure necessary for diverse recommendation). Our work establishes PRC as a new and complementary failure mode to neural collapse, specific to the deployment setting of compressed LLM-based recommenders.

### C. Practical Implications

For practitioners deploying LLM-based recommenders, we offer the following guidance. CARR achieves the best HR@10 among all LLM compression methods on ML-1M ($+8.3\%$ over Full-LLM) while preserving equivalent evidence-survival scores; monitor $\widehat{S}$ alongside HR/NDCG, especially for large catalogues where floor effects suppress HR@10. Avoid fixed-depth compression: CARR's adaptive depth selection identifies high collapse-risk conditions and applies conservative compression accordingly. CARR is strongest for cold-start items (HR@10$= 0.0043$ vs. zero for all other methods) and multi-intent users; for sparse or single-intent users, simpler fixed-compression methods are competitive.

### D. Limitations

Several limitations should be noted.

**Metric scale.** All LLM-based models are trained from scratch, so absolute HR@10 values are small (0.003–0.032 range), reflecting full-catalogue evaluation without negative sampling. The SGI component requires textual item metadata to pre-compute semantic cluster centroids; datasets without such metadata would need an alternative geometry-seeding mechanism.

**T5-Large evaluation.** Fine-tuning T5-Large (770M parameters) requires multi-GPU infrastructure beyond our current environment (Tesla T4, 16 GB VRAM); results are deferred to a future revision. The theoretical framework is architecture-agnostic.

**Component-level ablation.** Component-level attribution, isolating the contributions of $\mathcal{L}_{\text{collapse}}$, $\mathcal{L}_{\text{evidence}}$, and adaptive-depth selection, requires experiments with pre-trained backbones where regularisation effects are distinguishable within realistic training horizons. These are deferred to future work.

**Jacobian validation.** Direct spectral norm measurement yields near-unity values, making strict pointwise contraction difficult to confirm. The weaker expected-contraction condition $\mathbb{E}[\|J_l\|_2] < 1$ (Remark 1) is sufficient for the results to hold in expectation; the evidence-survival ordering in Table V provides the operative indirect support.

**Catalogue-scale robustness.** Experiment 10 (Steam, $281{,}428$ users) provides a first large-scale confirmation; scaling to $|\mathcal{I}| \in \{10^5, 10^6\}$ remains an open empirical task.

## XI. Conclusion

This paper formulated *Progressive Reasoning Collapse* (PRC) as a fundamental structural failure mode in compressed LLM-based generative recommendation. We introduced a formal framework capturing layerwise collapse of latent multi-intent geometry and decay of informative-history influence, and proved three theoretical results: monotonic collapse (Theorem 1), geometric evidence decay (Theorem 2), and the existence of a critical compression depth (Theorem 3).

CARR translates these insights into a collapse-aware compression method with adaptive depth and register-width selection. Experiments on five benchmarks against 12 baselines confirm the theoretical predictions: CARR achieves the highest HR@10 among all LLM compression methods across all five datasets, preserves evidence-survival scores equivalent to the uncompressed Full-LLM, and uniquely achieves non-zero HR@10 on 18,157 cold-start items where all other methods fail. These findings hold in a scratch-trained proof-of-concept setting; future work with pre-trained backbones is expected to amplify the observed advantages.

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**John Kingsley Arthur** received the Ph.D. degree in Computer Applied Technology from Jiangsu University, China, and is currently pursuing the Master of Professional Studies in Analytics (Applied Machine Intelligence) at Northeastern University, Toronto, Canada. His research interests include recommender systems, neural network optimization, large language models, and agentic artificial intelligence. He has authored and co-authored several peer-reviewed publications in international journals and conferences.

**Yvonne Leung** received the Ph.D. degree in Kinesiology and Health Science and a Doctoral Diploma in Health Psychology from York University, Toronto, ON, Canada. She held postdoctoral fellowships at Princess Margaret Cancer Centre and the University of Toronto, and served as an Assistant Professor in the Department of Psychiatry, Faculty of Medicine, University of Toronto. She is currently an Assistant Teaching Professor in the Master of Professional Studies in Analytics program at Northeastern University, Toronto. Her research interests include machine learning, NLP, and public health analytics.

**Vivian Clements Edwin** received the Ph.D. degree in Statistics from Bharathiar University, India. He is a Data Science Manager at Deloitte, Toronto, ON, Canada, and a Part-time Lecturer at Northeastern University, Toronto. With over 14 years of experience in enterprise analytics and ML, his research interests include deep learning, NLP, statistical modeling, and MLOps.

**Kofi Sarpong Adu-Manu** received the Ph.D. degree in computer science from the University of Ghana, Legon, Ghana. He is currently an Associate Professor with the Department of Computer Science, University of Ghana, and a Lead Instructor in AI at the Pan-African Doctoral Academy. His research interests include artificial intelligence, machine learning, wireless sensor networks, IoT, and cybersecurity.

**Kwabena Adu Agyemang** is a Ph.D. student at New Mexico State University, NM, USA. He holds the M.S. degree in Computer Science from New Mexico State University and the B.S. degree from Valley View University, Ghana. His research interests include AI in education, human-computer interaction, and emerging computing paradigms.